Programming Assignment Report

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Summary

The goal of this programming assignment is to evaluate two separate information retrieval (IR) methods and evaluate the performance of each method. The dataset for this assignment is a zip folder that contains 167,034 documents, with each document representing a test patient's clinical note with about 3,000-6,000 characters. Some examples of the information included within each document are the patient's ID, note title, clinical information, age, and gender. Two separate vector space model IR methods will be used, and each method will be evaluated using precision and recall.

Methods

To produce accurate outputs without errors, there were several pre-processing steps and packages used to clean the data. The main packages used include NLTK, Scikit-learn for the TF-IDF method, and Rankbm-25 for the second IR method. In addition to class materials, the Python Standard Library was referenced, and ChatGPT was used to troubleshoot errors. All text data was converted to lowercase, which is important to ensure the same word with different letters capitalized is evaluated as the same term (e.g., patient and Patient) (He, 2026a). Additionally, punctuation and stop-words were removed using the NLTK English stop-word list to remove words that are frequently used in our natural language, but they don't have much meaning (e.g., an, the, or, with) (He, 2026a). Word lemmatization was also used to reduce each word to its root word, such as reducing "patients" to "patient" (Nadkarni et al., 2011). Lemmatization helps the goal of our project to group each word's variations into a single term; for example, counting "patients" and "patient" as the same term (Manning et al., 2009). Performing these pre-processing steps helps to remove what's not important and distinguish the meaningful terms to find similar patients.

In Task 1, a vectorizer was used to build the TF-IDF matrices with several parameters. Smoothing was used (smoothidf=True) to prevent zero divisions, which gives some probability to words that are absent (Manning et al., 2009). Sublinear term frequency scaling was applied (sublinear_tf=True) to reduce the weight of words repeated frequently within a document, because it is unlikely that if the same word appears thirty times in a document, that it is thirty times more important than if it was mentioned one time (Manning et al., 2009). Then, the Euclidean distance, or L_2 distance, (norm="l2") normalization was applied so that documents that were longer didn't automatically seem more relevant due to having an increased number of terms (Manning et al., 2009). The

complete output matrix file was too large to load into GitHub, so a smaller matrix was also created into a .csv file so I could view it without using Python.

In Task 2, the other IR method chosen was Best Match 25 (BM25) using the rank_bm25 package with the Okapi BM25 (BM250kapi) algorithm. Both BM25 and TF-IDF are vector space models, with rankings based on the similarity measurement between each vector in space (Sivarajkumar et al., 2024). TF-IDF cosign similarity was used to identify the top five most similar documents. Okapi BM25 compares each term and the number of occurrences within a document with its frequency of the corpus. Unlike TF-IDF, BM25 takes into account the frequency of terms, the length of documents, and the average document length (Sivarajkumar et al., 2024). With the complete data set (opposed to the initial dataset file), BM25 took significantly longer, so a subset was used. A handful of different subset sizes were experimented with until ultimately deciding on 100 patients.

Vector space models can effectively rank document relevance; however, there are limitations. TF-IDF and BM25 do not understand the context of terms or how terms or documents relate to other documents within the corpus. For example, they cannot differentiate the same abbreviation with different meanings, depending on the context (e.g., “MAC” can stand for *monitored anesthesia care* or *mycobacterium avium complex*). Additionally, these methods do not understand or account for medical synonyms, such as “Emergency Department,” “Emergency Room”, “ED,” and “ER.”

In Task 3, we used precision and recall to qualitatively evaluate the IR methods. Precision @ K, with K=5 measures the precision for the top 5 retrieved documents. Precision is calculated by taking the number of relevant documents in the top K divided by K (He, 2026b). Recall @ K, with K=5 measures the recall for the top 5 retrieved documents. Recall is calculated by taking the number of relevant documents in the top K divided by the total number of relevant documents (He, 2026b). Task 3 was limited to only 100 patients because the dataset for Task 2 was limited to 100 patients, which causes a limitation when evaluating the results, as the complete corpus was not used.

Results

When calculating precision and recall, numbers that are higher, or closer, to 1.0 represent a better score. For the methods used, TF-IDF and BM25 performed almost the same. The average precision @ 5 for TF-TDF was 0.2360 and for BM25 it was 0.2280. The average recall @ 5 for BM25 for TF-IDF was 0.4485 and the average for BM25 was 0.4525. Recall scored higher than precision for both IR methods, showing almost half of relevant documents were found. These averages are represented in the table below.

Evaluation Method	TF-IDF	BM25
Average Precision @ 5	0.2360	0.2280
Average Recall @ 5	0.4485	0.4525

Because precision is evaluating if the IR method is actually retrieving relevant results, a higher precision number represents fewer false positive results, meaning fewer irrelevant results (He, 2026c). A lower precision result means that many of the documents that were retrieved are actually irrelevant. Recall results that are low numbers, or further away from 1.0, mean that the method is missing many relevant items (He, 2026c).

There appears to be a decent amount of overlap for the top five results with each IR method used. I believe this is consistent with the average numbers represented in the table above, given TF-IDF and BM25 performed very similarly. Below are a couple of examples of the top five results and each IR method with the colored boxes indicating overlapping results (these were randomly chosen while glancing at the Task 2 output file).

Patient_id: 88468, “1192821-2"	
TF-IDF	**BM25**
1192821-1	1192821-1
1764004-1	1764004-1
5412740-1	5412740-1
3713602-3	3713602-3
3713602-1	3713602-1

Patient_id: 88482, “1208903-1"	
TF-IDF	**BM25**
2771962-2	2771962-1
4128839-1	2771962-2
4128839-4	6611245-1
2771962-1	5203866-1
4128839-3	7236174-1

Patient_id: 88483, “1208912-1"	
TF-IDF	**BM25**
3466823-1	4881680-1
4881680-1	3466823-1
5779145-1	3466899-1
2918532-1	2918532-1
3391625-1	7203858-1

References

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